

Shekh. Md. Saifur Rahman

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RESEARCH INTEREST

My research aim is to build **accessible AI and Computer Vision** systems for real-world analysis and interaction, addressing critical challenges in environmental sustainability and the preservation of our cultural legacies.

EDUCATION

- **Military Institute of Science and Technology** Jan 2020 - Mar 2024
B.SC. IN COMPUTER SCIENCE AND ENGINEERING Dhaka, Bangladesh
 - CGPA: 3.86/4.00
 - Thesis Title: PersiDani Fusion: Investigating Jamdani and Persian Motif Fusion through Image Processing, Transfer Learning, and GAN

RESEARCH EXPERIENCE

UIU Advanced Underwater Robotics & Automation (AURA) Crew, UIU

- Role: **Mentor** | Status: **Ongoing** | Feb 2025 - Present
- Mentoring the UIU AURA Crew in the development of advanced Remotely Operated Vehicles (ROVs) capable of performing complex underwater tasks beyond human limitations, focusing on robotic design, sensor integration, and control systems.
- Led **UIU MARINER**, the first Bangladeshi team to qualify for the **MATE ROV World Championship 2025** (USA), fostering innovation, system prototyping, and cross-disciplinary collaboration.
- Advisor: [Dr. M. Rezwana Khan](#), [Dr. A.K.M. Muzahidul Islam](#).

PersiDani Fusion: Investigating Jamdani and Persian Motif Fusion through Image Processing, Transfer Learning, and GAN | Status: Completed

- **Undergraduate Thesis** | Supervised by [Nuzhat Tabassum](#). The research has been submitted to *Neural Computing and Applications* for publication. [\[Thesis Details\]](#)
- Explored the fusion of traditional Jamdani and Persian motifs by applying image processing, transfer learning, and generative adversarial networks (GANs) to generate culturally rich textile patterns.
- Developed a custom Persian motif dataset and utilized StyleGAN to synthesize novel designs, followed by a comparative analysis of generated motifs to assess aesthetic and cultural coherence, contributing to textile design and pattern synthesis.

Text to Circuit: Multi-Model Approaches (GANs, Diffusion, VAEs) for Accurate Electrical Diagram Generation

- Role: **Research Lead** | Status: **Ongoing** | July 2025 - Present
- Leading a research project aimed at overcoming the limitations of state-of-the-art text-to-image models in generating topologically accurate technical diagrams from complex natural language prompts.
- Directing the design, development, and training of multiple generative architectures (GANs, Diffusion Models, and VAEs) to produce high-fidelity circuit diagrams.
- Focusing on a model that can understand a prompt's technical details to generate circuit diagrams that are both visually accurate and logically correct.

IoT Networking & Data Communication Repository

- Role: **Research Collaborator** | Status: **Ongoing** | Apr 2025 - Present
- Collaborating on the development of **NetIoT**, an open-source repository offering hands-on tutorials and real-world projects focused on IoT systems, networking, and data communication using Arduino, ESP32, and Raspberry Pi.
- Contributing to modules on sensor integration, automation, GUI development, and remote communication to help learners build scalable IoT-based applications.
- Advisor: [Mohammad Mamun Elahi](#)

MIST Mars Rover Society, MIST

- Role: **Co-lead – Software & Autonomous Navigation Team; Member – Science Team** | Oct 2021 - Mar 2024
- Led the design and development of autonomous navigation systems, including path planning, obstacle avoidance, and localization for Mars rover prototypes in simulation and real-world field tests.
- Worked on the science module by integrating and calibrating various soil-testing sensors for environmental data collection and terrain analysis.
- Contributed to the team's success in international competitions, achieving **Champion at URC 2021, 2nd Runner-up at ARC 2022, and 5th place at ARC 2023**.

PUBLICATIONS

J = JOURNAL, C = CONFERENCE

Journal

- [J.1] Rahman, S.M.S., Fahmida, N.N., Khan, E.A., Zerin, S.N., Tabassum, N. and Sharmin, N., 2025. PersiDani Fusion: Investigating Jamdani and Persian Motif Fusion through Image Processing, Transfer Learning, and GAN. In *Neural Computing and Applications* (Manuscript submitted). 📄

Conference

- [C.1] Orchi, A.T., Zaman, M.F., Khan, M.D., Alamgir, M., Islam, M., Rahman, M.J., Abdullah, A.M.Z., Khan, M.M.H., Baki, M.K.A., Repat, I.L., Ahmed, M.S., Hossain, M.S., Islam, M., Seaman, S.M., Piyal, N.J., **Rahman, S.M.S.**, Hafiz, F., Islam, A.K.M.M., and Khan, M.R., 2026. HyDRA Scorpion: An AI-Driven Modular ROV for Real-Time Underwater Inspection, Intervention, and Object Detection. In *2026 IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, June 1–5* (Manuscript submitted). 
- [C.2] Bilwal, M., Tanvin, J.U., **Rahman, S.M.S.**, Faisal, M.R.F., Khan, A.F., Arafat, S.E. and Akhtaruzzaman, M., 2025. A Multi-Modal Assistive Robot Navigation System for Physically Impaired Users. In *2025 IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (ICQPAIN)* (Manuscript presented). 
- [C.3] Sabir, N., Sajjad, S.M., Mahmood, M., Ahmed, M.R., Hassan, M.R., and **Rahman, S.M.S.**, 2025. Vision-Based Traffic Signal Automation and Parking System with Emergency Vehicle Prioritization. In *2025 International Conference on Computer and Information Technology (ICCIT)* (Manuscript submitted). 
- [C.4] Zerin, M., **Rahman, S.M.S.**, Fahmida, N.N., Zerin, S.N. and Islam, M.N., 2025, February. Evaluating and Developing Educational Gaming Applications to Raise Awareness of Child Sexual Abuse. In *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)* (pp. 1-6). IEEE. 

PROFESSIONAL EXPERIENCE

- **Lecturer**, Department of Computer Science and Engineering Oct 2024 - present
United International University, Dhaka, Bangladesh
 - **Course Taught:** Data Structures and Algorithms - I; Data Structures and Algorithms - II; Artificial Intelligence; Database Management System; Object Oriented Programming; Theory of Computation; Microprocessors and Microcontrollers; Electrical Circuits; Society, Environment, and Engineering Ethics.
 - **Provided technical and strategic mentorship** to software and hardware project groups across multiple courses; One of the mentored teams secured 2nd position in a project showcase at UIU.
 - **Prepared lectures and class activities**, integrated multimedia approaches, and used instructional technology to enhance the pedagogical approach following an OBE-based curriculum.
 - **Conducted Sessions on Research Workflow:** Research ideation, literature review, and research methodologies, guiding students to develop a strong foundation for their research projects.
- **Team Mentor**, UIU Advanced Underwater Robotics & Automation Crew Feb 2025 - present
United International University, Dhaka, Bangladesh
 - **Mentoring a multidisciplinary team** to design and develop an underwater robot for international robotics competitions, focusing on embedded systems, control logic, and systems integration.
- **Lecturer**, Department of Computer Science and Engineering Apr 2024 - Oct 2024
Military Institute of Science and Technology, Dhaka, Bangladesh
 - **Course Taught:** Data Structures and Algorithms - I
- **Software Engineer Intern**, IT Division Jan 2023 – Feb 2023
Trust Bank PLC., Dhaka, Bangladesh 
 - **Developed a comprehensive trainee management system** for Trust Bank, streamlining trainee tracking, evaluation, and administrative processes.
 - **Technology:** MySQL, PHP, JavaScript, HTML, CSS, Bootstrap

CERTIFICATIONS

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| Oct 2023 | Fundamental IT Engineer Examinations, ITEE
Information Technology Promotion Agency, Japan (IPA) | [Certificate] |
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SELECTED PROJECTS

- **NextGen Wheelchair** Sep 2022 - Jun 2023
Technology: Machine Learning, Image Processing, Python, Arduino, JavaScript, IoT  
 - Engineered an IoT-enabled, multi-modal wheelchair system integrating real-time visual processing and sensor-based control for remote navigation, aimed at enhancing mobility for physically impaired users. The project focused on adaptive control logic and usability in constrained environments.
- **Heatmap Generation from Feet Pressure** Dec 2022 - Feb 2023
Technology: Python, Arduino, Pressure Sensors, IoT 
 - Designed a wearable IoT system using pressure sensors to capture plantar pressure distribution and generate dynamic heatmaps. The project aimed to support gait analysis and orthopedic assessment, with potential use in fall detection and rehabilitation monitoring.

- **Bangla Book Recommendation System** Mar 2023 – Oct 2023
Technology: Python, Pandas, NumPy, Scikit-learn
 ◦ Designed and implemented a collaborative filtering-based recommendation system tailored for Bangla literature readers, using anonymized user preference data. Explored user similarity metrics and evaluated system performance on a sparse dataset.
- **Educational Gaming for CSA Awareness (Research Publication)** Mar 2023 – Feb 2025
Technology: Usability Evaluation, Heuristic Evaluation, SUS 
 ◦ Co-authored a research project accepted at ECCE 2025, involving the design and development of a Bengali storytelling game aimed at educating children about child sexual abuse (CSA) through interactive scenarios. Conducted heuristic evaluations and System Usability Scale (SUS) assessments to validate its effectiveness and usability.

AWARDS & HONORS

2020 - 2024	Commandant's List of Honour Military Institute of Science and Technology, Dhaka, Bangladesh ◦ Achieved a GPA of 3.80/4.00 or above in all four academic years	
	Dean's List of Honour Military Institute of Science and Technology, Dhaka, Bangladesh ◦ Achieved a GPA of 3.70/4.00 or above in all four academic years	[Certificate]
2023	Champion in the Independence Day Programming Contest (IDPC) 2023 Military Institute of Science and Technology, Dhaka, Bangladesh	[Certificate]
	5th in the Anatolian Rover Challenge 2023 Space Exploration Society(UKET), Turkiye	[Certificate]
	44th in the SUST Inter University Programming Contest (IUPC) 2023 Shahjalal University of Science and Technology, Sylhet, Bangladesh	[Certificate]
2022	2nd Runner-up in the Anatolian Rover Challenge 2022 Space Exploration Society(UKET), Turkiye	
	43rd in the ICPC Asia Dhaka Regional Contest 2022 ICPC, Asia West Region – Dhaka Site	[Certificate]
2021	Champion in the University Rover Challenge (URC) 2021 The Mars Society, USA	[Certificate]

EXTRACURRICULAR & LEADERSHIP EXPERIENCE

Jul 2021 - Mar 2024	Data Structures and Algorithms Instructor , MIST Computer Club Military Institute of Science and Technology, Dhaka, Bangladesh
Feb 2023 - Mar 2024	Vice President, Committee of R&D , MIST Computer Club Military Institute of Science and Technology, Dhaka, Bangladesh
Oct 2021 - Mar 2024	Software Team Member , MIST Mars Rover Society Military Institute of Science and Technology, Dhaka, Bangladesh • Designed and implemented an autonomous navigation system for a Mars rover
Apr 2021 - Mar 2022	Mentor, Technical team , MIST Robotics Club Military Institute of Science and Technology, Dhaka, Bangladesh
Apr 2022 - Aug 2022	Executive Director , MIST Robotics Club Military Institute of Science and Technology, Dhaka, Bangladesh

SKILLS & INTERESTS

Programming Languages: C/C++, Java, Python, PHP, JavaScript, SQL, HTML, CSS, Bash
Frameworks & Libraries: Django, Flask, React, .NET, Bootstrap, TensorFlow, Keras, PyTorch
Tools & Platforms: Linux, Git, MATLAB, Arduino IDE, Cisco Packet Tracer, Proteus
Hobbies: Playing Badminton, Carrom, and Table Tennis

REFERENCES

Dr. A.K.M. Muzahidul Islam

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Department of CSE, UIU

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Dr. Muhammad Nazrul Islam

Associate Professor
Department of CSE, MIST

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Nuzhat Tabassum (Thesis Supervisor)

Assistant Professor
Department of CS, AIUB

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